

**GIRRAWEE HIGH SCHOOL
MATHEMATICS**

Year 12 Extension 2 Task 34?

Monday 21st June 2004

- Instructions: a) Write all your answers on your own paper.
b) Show all necessary working.
c) Marks may be deducted for careless or badly arranged work.

Time Allowed: 90 minutes

Question 1 (28 marks)

Marks

a) Find the following integrals:

- | | |
|---|---|
| (i) $\int \sin^3 2x dx$ | 4 |
| (ii) $\int x^2 e^{2x} dx$ | 4 |
| (iii) $\int \frac{x^2 dx}{\sqrt{(1-x^2)^3}}$ | 4 |
| (iv) $\int_0^{\frac{2\pi}{3}} \frac{1}{5+4\cos x} dx$ | 4 |
| (v) $\int \frac{x}{\sqrt{1-2x-x^2}} dx$ | 4 |
| (vi) $\int \sec^3 x dx$ | 4 |
| (vii) $\int \frac{2x+3}{(x-2)(x^2+3)} dx$ | 4 |

Question 2 (11 marks)

- | | |
|---|---|
| a) (i) If $I_n = \int_1^e x(\log x)^n dx$ for $n \geq 0$ show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$ for $n \geq 1$ | 3 |
| (ii) Evaluate $\int_1^e x(\log x)^3 dx$ | 3 |
| b) (i) Use the substitution $u = \frac{1}{x}$, show that $\int_{0.5}^1 \frac{\log x}{1+x^2} dx = \int_2^1 \frac{\log u}{1+u^2} du$ | 3 |
| (ii) Hence, or otherwise, evaluate $\int_{0.5}^2 \frac{\log x}{1+x^2} dx$ | 2 |

Question 3 (25 marks)

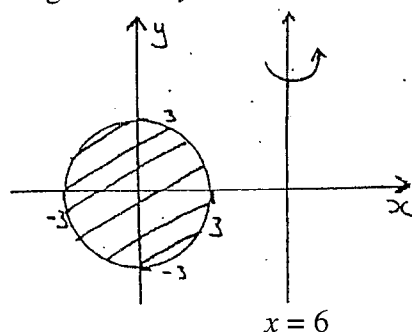
Marks

- a) The area bounded by the curve $y = x^2 + 1$ and the line $y = 3 - x$ is rotated about the x -axis.

(i) Sketch the curve and line clearly labelling all the points of intersection. 3

(ii) By considering slices perpendicular to the x -axis, find the volume of the solid formed. 5

- b) The diagram shows the region $x^2 + y^2 \leq 9$ and the line $x = 6$

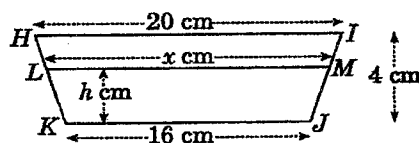


- (i) Use the method of cylindrical shells to show that if the region $x^2 + y^2 \leq 9$ is rotated about the line $x = 6$, the volume V of the torus formed is given by; 3

$$V = 4\pi \int_{-3}^3 (6-x)\sqrt{9-x^2} dx$$

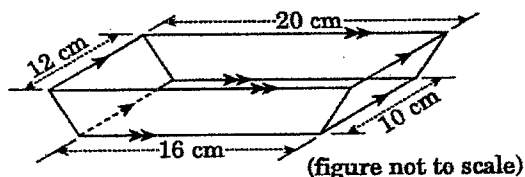
(ii) Hence find the volume of the torus. 3

- c) A solid has a base in the form of an ellipse with major axis 10 units and minor axis 8 units. If every section perpendicular to the major axis is an isosceles triangle with altitude 6 units, find the volume of the solid. 5



d)

- (i) A trapezium $HIJK$ has parallel sides $KJ = 16$ cm and $HI = 20$ cm. The distance between these sides is 4 cm. L lies in HK and M lies on IJ such that LM is parallel to KJ . The shortest distance from K to LM is h cm and LM has length x cm. Prove that $x = 16 + h$. 2



(ii) (figure not to scale) 4

The diagram above is of a cake tin with a rectangular base with sides 16 cm and 10 cm. Its top is also rectangular with dimensions 20 cm and 12 cm. The tin has depth 4 cm and each of its four side faces is a trapezium. Find its volume.

Question 1 (28)

a) i) $\int \sin^3 2x \, dx$

$$= \int \sin 2x (1 - \cos^2 2x) \, dx$$

$$= -\frac{1}{2} \int (1 - u^2) \, du \quad \begin{matrix} u = \cos 2x \\ du = -2 \sin 2x \, dx \end{matrix}$$

$$= -\frac{1}{2} (u - \frac{1}{3} u^3) + c$$

$$= -\frac{1}{2} \cos 2x + \frac{1}{6} \cos^3 2x + c \quad (4)$$

ii) $\int x^2 e^{2x} \, dx \quad u = x^2 \quad v = \frac{1}{2} e^{2x}$

$$= \frac{1}{2} x^2 e^{2x} - \int x e^{2x} \, dx$$

$$= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x} + c \quad (4)$$

iii) $\int \frac{x^2 \, dx}{\sqrt{1-x^2}}$ $x = \sin \theta \quad dx = \cos \theta \, d\theta$

$$= \int \frac{\sin^2 \theta \cos \theta \, d\theta}{\cos^3 \theta}$$

$$= \int \tan^2 \theta \, d\theta$$

$$= \int (\sec^2 \theta - 1) \, d\theta$$

$$= \tan \theta - \theta + c$$

$$= \frac{x}{\sqrt{1-x^2}} - \sin^{-1} x + c \quad (4)$$

iv) $\int_0^{\frac{\pi}{3}} \frac{1}{5+4\cos x} \, dx \quad t = \tan \frac{x}{2}$

$$= \int_0^{\frac{\pi}{3}} \frac{2 \, dt}{5+4(\frac{1-t^2}{1+t^2})}$$

$$= \int_0^{\frac{\pi}{3}} \frac{2 \, dt}{9+t^2}$$

$$= \frac{2}{3} \left[\tan^{-1} \frac{t}{3} \right]_0^{\frac{\pi}{3}}$$

$$= \frac{2}{3} \left\{ \tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right\}$$

$$= \frac{2}{3} \times \frac{\pi}{6}$$

$$= \frac{\pi}{9} \quad (4)$$

v) $\int \frac{x \, dx}{\sqrt{1-2x-x^2}}$

$$= -\frac{1}{2} \int \frac{-2x-2}{\sqrt{1-2x-x^2}} \, dx = \int \frac{dx}{\sqrt{2-(x+1)^2}}$$

$$= -\sqrt{1-2x-x^2} - \sin^{-1} \left(\frac{x+1}{\sqrt{2}} \right) + c \quad (4)$$

vi) $\int \sec^3 x \, dx = \int \sec x \sec^2 x \, dx$

$$u = \sec x \quad v = \tan x$$

$$du = \sec x \tan x \, dx \quad dv = \sec^2 x \, dx$$

$$= \sec x \tan x - \int \sec x \tan^2 x \, dx$$

$$= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$$

$$\therefore 2 \int \sec^3 x \, dx = \sec x \tan x + \log(\sec x + \tan x) + c$$

$$\int \sec^3 x \, dx = \frac{1}{2} (\sec x \tan x + \log(\sec x + \tan x)) + c \quad (4)$$

vii) $\frac{A}{x-2} + \frac{Bx+C}{x^2+3} = \frac{2x+3}{(x-2)(x^2+3)}$

$$A(x^2+3) + (Bx+C)(x-2) = 2x+3$$

$$\begin{matrix} x=2 & x=0 & x=-1 \\ 7A=7 & 3A-2C=3 & 4A-B-C=5 \\ A=1 & 3-2C=3 & 4-B-C=5 \\ & C=0 & B=-1 \end{matrix}$$

$$\int \frac{2x+3}{(x-2)(x^2+3)} \, dx = \int \left[\frac{1}{x-2} - \frac{x}{x^2+3} \right] \, dx$$

$$= \log(x-2) - \frac{1}{2} \log(x^2+3) + c \quad (4)$$

Question 2 (11)

i) a) $I_n = \int_1^e x (\log x)^n \, dx$

$$u = (\log x)^n \quad v = \frac{1}{2} x^2$$

$$du = n(\log x)^{n-1} \frac{1}{x} \, dx \quad dv = x \, dx$$

$$I_n = \left[\frac{1}{2} x^2 (\log x)^n \right]_1^e - \frac{n}{2} \int_1^e x (\log x)^{n-1} \, dx$$

$$= \frac{e^2}{2} - \frac{n}{2} I_{n-1} \quad (3)$$

iii) $\int_1^e x (\log x)^3 \, dx = I_3$

$$= \frac{e^2}{2} - \frac{3}{2} I_2$$

$$= \frac{e^2}{2} - \frac{3e^2}{4} + \frac{3}{2} I_1$$

$$= \frac{e^2}{4} + \frac{3e^2}{4} - \frac{3}{4} I_0$$

$$= \frac{e^2}{2} - \frac{3}{4} \int_1^e x \, dx$$

$$= \frac{e^2}{2} - \frac{3}{4} \left[\frac{1}{2} x^2 \right]_1^e$$

$$= \frac{e^2}{2} - \frac{3}{8} e^2 + \frac{3}{8}$$

$$= \frac{e^2+3}{8} \quad (3)$$

b) i) $\int_{0.5}^1 \frac{\log x}{1+x^2} \, dx \quad u = \frac{1}{x} \Rightarrow x = \frac{1}{u}$

$$dx = -\frac{du}{u^2}$$

$$= \int_2^1 \frac{-\log(\frac{1}{u})}{1+(\frac{1}{u^2})} \frac{du}{u^2}$$

$$= \int_2^1 \frac{\log u}{u^2+1} \, du \quad (3)$$

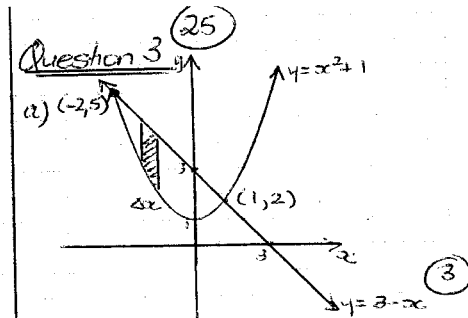
ii) $\int_{0.5}^2 \frac{\log x}{1+x^2} \, dx$

$$= \int_{0.5}^1 \frac{\log x}{1+x^2} \, dx + \int_1^2 \frac{\log x}{1+x^2} \, dx$$

$$= \int_2^1 \frac{\log x}{1+x^2} \, dx + \int_1^2 \frac{\log x}{1+x^2} \, dx$$

$$= -\int_1^2 \frac{\log x}{1+x^2} \, dx + \int_1^2 \frac{\log x}{1+x^2} \, dx$$

$$= 0 \quad (2)$$



$$x^2 + 1 = 3 - x$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2 \text{ or } x = 1$$

ii) $A(x) = \pi[(3-x)^2 - (x^2+1)^2]$

$$\Delta V = \pi(9 - 6x + x^2 - x^4 - 2x^2 - 1)$$

$$= \pi(8 - 6x - x^2 - x^4) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=2}^1 \pi(8 - 6x - x^2 - x^4) \Delta x$$

$$= \pi \int_2^1 (8 - 6x - x^2 - x^4) \, dx$$

$$= \pi \left[8x - 3x^2 - \frac{1}{3} x^3 - \frac{1}{5} x^5 \right]_2^1$$

$$= \pi \left\{ \left(8 - 3 - \frac{1}{3} - \frac{1}{5} \right) - \left(16 - 12 + \frac{8}{3} + \frac{32}{5} \right) \right\}$$

$$= \frac{117\pi}{5} \text{ units}^3 \quad (5)$$

b) $2y = 2\sqrt{9-x^2}$

$$A(x) = 4\pi(6-x)\sqrt{9-x^2}$$

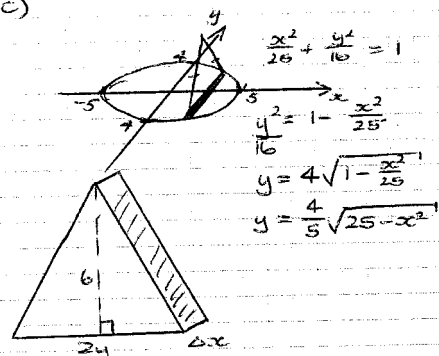
$$\Delta V = 4\pi(6-x)\sqrt{9-x^2} \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=3}^0 4\pi(6-x)\sqrt{9-x^2} \Delta x$$

$$= 4\pi \int_3^0 (6-x)\sqrt{9-x^2} \, dx \quad (3)$$

$$\begin{aligned}
 \text{(ii)} \quad V &= 4\pi \int_{-3}^3 (6-x)\sqrt{9-x^2} dx \\
 &= 24\pi \int_{-3}^3 \sqrt{9-x^2} dx - 4\pi \int_{-3}^3 x\sqrt{9-x^2} dx \\
 &\quad \begin{array}{cc} \uparrow & \uparrow \\ \text{semicircle} & \text{odd} \times \text{even} \\ & = \text{odd function} \end{array} \\
 &= 24\pi \times \frac{1}{2} \pi (3)^2 - 0 \\
 &= \underline{108\pi^2 \text{ units}^3} \quad (3)
 \end{aligned}$$

c)



$$= \frac{8}{5}\sqrt{25-x^2}$$

$$\begin{aligned}
 A(x) &= \frac{1}{2} \times \frac{8}{5}\sqrt{25-x^2} \times 6 \\
 &= \frac{24}{5}\sqrt{25-x^2}
 \end{aligned}$$

$$\Delta V = \frac{24}{5}\sqrt{25-x^2} \Delta x$$

$$\begin{aligned}
 V &= \lim_{\Delta x \rightarrow 0} \sum_{x=-5}^5 \frac{24}{5}\sqrt{25-x^2} \Delta x \\
 &= \frac{24}{5} \int_{-5}^5 \sqrt{25-x^2} dx
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{24}{5} \times \frac{1}{2} \pi (5)^2 \\
 &= \underline{60\pi \text{ units}^3} \quad (5)
 \end{aligned}$$

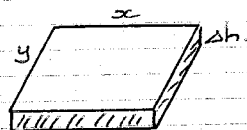
$$d) m = \frac{4}{20-16} = 1$$

$$\begin{aligned}
 h-10 &= 1(x-16) \\
 &= x-16 \\
 x &= h+16 \quad (2)
 \end{aligned}$$

(ii) y relationship

$$\begin{aligned}
 m &= \frac{4}{12-10} \\
 &= 2.
 \end{aligned}$$

$$\begin{aligned}
 h-0 &= 2(y-10) \\
 h &= 2y-20 \\
 y &= \frac{1}{2}h+10
 \end{aligned}$$



$$\begin{aligned}
 A(h) &= (h+16)(\frac{1}{2}h+10) \\
 &= \frac{1}{2}h^2 + 18h + 160
 \end{aligned}$$

$$\Delta V = (\frac{1}{2}h^2 + 18h + 160) \Delta h$$

$$V = \lim_{\Delta h \rightarrow 0} \sum_{h=0}^4 (\frac{1}{2}h^2 + 18h + 160) \Delta h$$

$$= \int_0^4 (\frac{1}{2}h^2 + 18h + 160) dh$$

$$= \left[\frac{1}{6}h^3 + 9h^2 + 160h \right]_0^4$$

$$= \frac{32}{3} + 144 + 640$$

$$= \underline{794 \frac{2}{3} \text{ units}^3} \quad (4)$$